

Available online at www.sciencedirect.com

Resuscitation Plus

journal homepage: www.elsevier.com/locate/resuscitation-plus

Clinical paper

Epidemiology and outcomes of out-of-hospital cardiac arrests treated by an anaesthetist-staffed emergency medical service: a 3-year registry analysis in The Friuli-Venezia-Giulia region



Carlo Pegani^a, Alberto Peratoner^a, Manila Andrian^b, Laura Magagnin^c, Luca Gambolò^d, Alice Pravisani^e, Michele Zuliani^f, Serena Rakar^g, Cristina Lutman^h, Alessio Della Mattiaⁱ, Katya Fabrettoⁱ, Annarita Tullio^j, Giuseppe Stirparo^d, Erik Roman Pognuz^k, Giuseppe Ristagno^{l,m,*}, the FVG-OHCA registry network¹

Abstract

Objectives: Out-of-hospital cardiac arrest (OHCA) presents significant regional variations in incidence, management, and survival rates. The Friuli-Venezia-Giulia (FVG) region in northeastern Italy has established a cardiac arrest registry to evaluate epidemiological trends and the effectiveness of its emergency medical service (EMS) interventions. This study analyses EMS-treated OHCA over a three-year period, focusing on patient characteristics, resuscitation practices, and survival outcomes.

Methods: A retrospective cohort study was conducted using prospectively collected data from the FVG-OHCA registry between January 2021–December 2023. All adult OHCA where resuscitation was attempted by EMS were included. Demographics, OHCA characteristics, bystander interventions, EMS response and treatments, and outcomes were analysed. Logistic regression was used to identify factors associated with survival to hospital admission, six-month survival, and good neurological recovery (Cerebral Performance Category (CPC) 1–2).

Results: A total of 4,089 OHCA cases were recorded, with an incidence of 113/100,000 inhabitants/year. Although bystander CPR rate was 67%, public AED use was low (4.3%). Resuscitation was attempted by EMS in 48% of cases, and an advanced airway was placed in 75% of patients. Survival to hospital admission was 22.9%, while six-month survival was 9.7%, and 7.6% of patients had a CPC 1–2. Younger age, male gender, shockable rhythm, and public location were associated with long-term survival. High-quality bystander CPR, use of mechanical CPR, and advanced airway placement during CPR were associated only with survival to hospital admission.

Conclusions: This study provides comprehensive insights into OHCA epidemiology and outcomes in the FVG region and emphasises the importance of early intervention, high-quality bystander CPR, and specialised prehospital care.

Keywords: Out-of-hospital cardiac arrest, Registry, Survival, Cardiopulmonary resuscitation, Outcome

Introduction

Out-of-hospital cardiac arrest (OHCA) is one of the leading health problems worldwide, with European incidence ranging from 67 to 170 cases per 100,000 inhabitants annually, and cardiopulmonary

resuscitation (CPR) attempted or continued by the emergency medical service (EMS) in approximately 50% of instances.¹ Despite advances in resuscitation science, survival to hospital discharge remains poor and varies significantly between regions and countries.^{1,2} The second report from the International Liaison Committee on Resuscitation (ILCOR), encompassing eleven national and four

* Corresponding author at: Department of Pathophysiology and Transplantation, University of Milan, Italy; Department of Anaesthesiology, Intensive Care and Emergency, Fondazione IRCCS Ca' Granda Ospedale Maggiore Policlinico, Via Della Commenda 16, 20122 Milan, Milan, Italy.

E-mail address: gristag@gmail.com (G. Ristagno).

¹ The members of the FVG-OHCA registry network are listed in Appendix 1 at the end of the article.

<https://doi.org/10.1016/j.resplu.2025.101000>

Received 4 March 2025; Received in revised form 31 May 2025; Accepted 4 June 2025

regional cardiac arrest registries across the world, reported survival rates ranging from 4.6% to 16% for EMS-treated OHCA.³

Significant efforts have sought to understand the reasons behind such disparities in outcomes across realities.¹ Key contributing factors include differences in data collection standards, case-mix, community engagement, EMS organisation, CPR quality, and post-resuscitation care.^{4,5} The initial minutes following collapse are particularly critical, with survival influenced heavily by actions taken by bystanders, first responders, and EMS personnel.⁶ Thus, EMS practice, including dispatcher-assisted (DA) bystander CPR, AED availability, ambulance crew composition, and provision of high-quality CPR, play a pivotal role in determining outcomes. To better understand these determinants for OHCA survival and benchmark EMS performance, several countries have implemented cardiac arrest registries.^{7,1–3}

Since 2021, the Friuli-Venezia-Giulia (FVG) region in northeastern Italy has established a cardiac arrest registry, providing a valuable opportunity for data collection and analysis. This region offers a unique context for studying OHCA due to its diverse geography, spanning from urban, rural, and mountainous areas, which influences EMS accessibility and response times (<https://www.regione.fvg.it/rafvfg/cms/RAFVFG/>).

This study aims to provide a comprehensive overview of OHCA cases with EMS-attempted resuscitation in the FVG region over a 3-year period. Specifically, we examined incidence rates, demographic and clinical characteristics, bystander and EMS interventions, and their association with survival and neurological recovery.

Methods

This retrospective cohort study was based on prospectively collected data from the FVG-OHCA registry, covering the period from January 1st 2021 to December 31st 2023. The study was approved by the Regional Ethics Committee of FVG (no. CEUR-2020-Os-168).

FVG emergency medical system

FVG Region covers an area of 7924 Km², is divided into 4 provinces (Triest, Udine, Pordenone, and Gorizia), and hosts a total of 1,201,513 inhabitants (Fig. 1). Provinces encompass diverse settings, including urban areas, countryside, and, particularly Udine, large alpine mountainous terrains. Population density varies across the provinces, depending on the territorial characteristics and the level of urbanization. EMS is coordinated by the “Struttura Operativa Regionale Emergenza Sanitaria” (SORES), located in Palmanova (Udine), which ensures standardised emergency medical procedures throughout the region.

The regional EMS comprises: 9 basic life support (BLS) vehicles, staffed by 3 volunteer rescuers trained in BLS and defibrillation; 42 advanced life support (ALS) vehicles, staffed by an emergency nurse and two trained rescuers; 7 ALS medical cars, staffed by an anaesthetist; and 1 helicopter EMS, staffed by a nurse flight coordinator and an anaesthetist. Emergency vehicle availability varies between day and night, as shown in Fig. 1. For OHCA events, EMS protocol includes simultaneous dispatch of an ALS nurse ambulance and an ALS medical car, except in the mountainous areas of Udine, where only the ALS nurse ambulance is deployed. Thus, EMS management of OHCA patients in the Region is organised with ALS teams led by anaesthetists. A mechanical chest compression device (LUCAS, Stryker, Sweden) is available on 20% of ALS vehicles in Trieste,

27% in Udine, 100% in Pordenone, and 42% in Gorizia. Post-cardiac arrest patients are generally transported to one of the 3 regional Hub hospitals (Fig. 1).

FVG-OHCA registry

Although approved in June 2020, data collection for the registry began in January 2021 due to the COVID-19 pandemics. All adult OHCA cases where EMS attempted resuscitation (i.e. cardiac arrests treated) are included in the registry. The registry was developed based on the Utstein Registry template.⁸ However, while the data collection format is broadly aligned with Utstein recommendations, not all Utstein-defined variables are included. Some variables have been adapted to the local context and available data sources, particularly for tracking long-term outcomes. Six-month survival and neurological outcome are determined either during cardiology follow-up or via telephone interviews with patients or their relatives. EMS staff enter data on a voluntary basis, and anonymized data are verified for accuracy by chief medical officers.

Statistical analyses

Data collected included: patients' characteristics (age, gender), OHCA feature (initial presenting rhythm, witnessed, bystander intervention, no-flow and low-flow duration), EMS intervention (use of public defibrillation access, DA-CPR, intervention time, use of mechanical chest compression device, drugs, airway management). “No-flow” duration was defined as the interval from emergency call dispatch to initiation of CPR (by either bystanders or EMS). “Low-flow” duration was defined as the interval from CPR initiation to either ROSC or termination of CPR in the instance of unsuccessful resuscitation. Bystander CPR quality was subjectively assessed by EMS personnel upon arrival at the scene and categorised as either “high” or “poor,” based on visible CPR characteristics (i.e. hands position, compression depth, compression rate, chest recoil, and interruptions).

Incidence rates were calculated using the population covered and extrapolated to incidence rates per 100,000/year. Descriptive analyses were performed for the entire region and each province. Categorical variables were presented as numbers and corresponding percentages, and continuous variables as medians and interquartile ranges (IQR). Differences between provinces were tested using Pearson's Chi-Square test for categorical variables and Kruskal-Wallis test for continuous variables.

Univariable and multivariable logistic regression models were applied to determine the association between cardiac arrest characteristics and EMS interventions and survival to hospital admission, 6-month survival, and 6-month good neurological recovery, defined as Cerebral Performance Category (CPC) 1–2. Absence of multicollinearity between independent factors was verified. Crude and adjusted odds ratios, with the corresponding 95% confidence interval, were reported.

All statistical analyses were performed using SPSS version 29 and SAS version 9.4. A p value < 0.05 was considered as statistically significant.

Results

A total of 4,089 OHCA cases were attended by EMS in the FVG region between 2021 and 2023, resulting in an overall incidence of 113 per 100,000 inhabitants per year (Fig. 2). In 2129 cases,

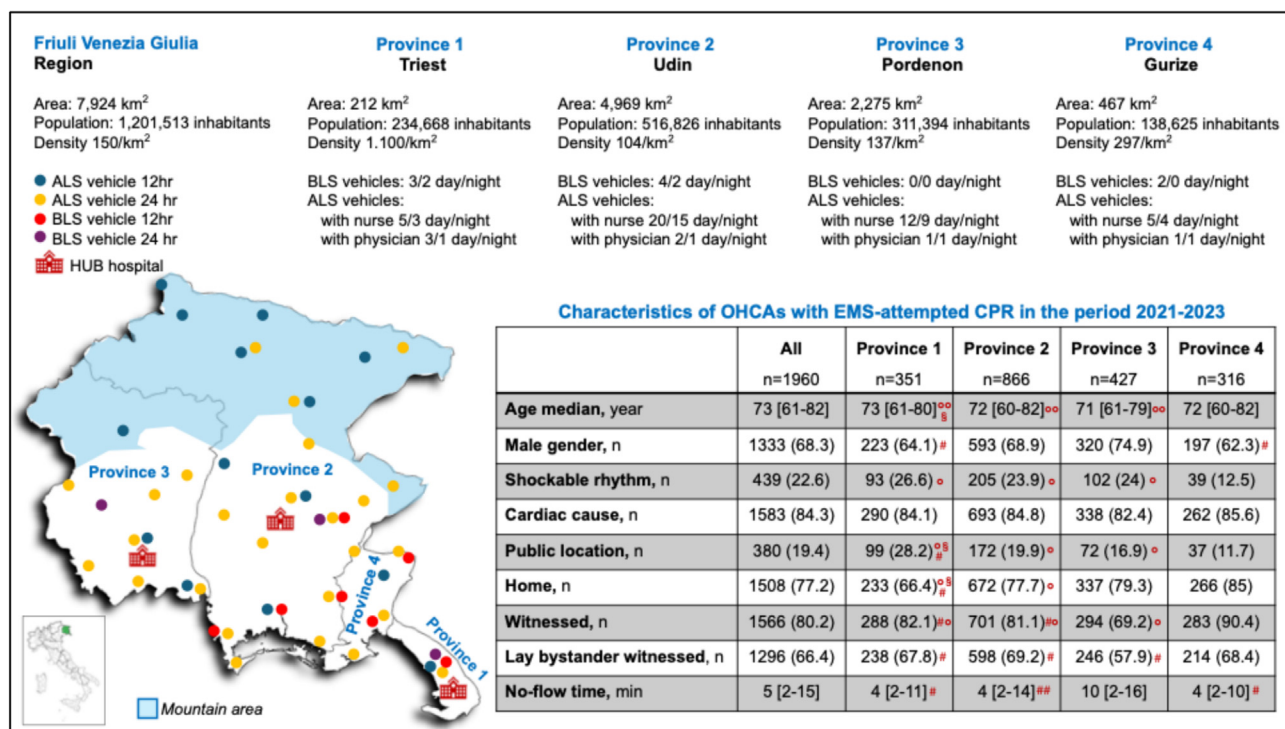


Fig. 1 – Demographic data and emergency medical system vehicles distribution and availability in Friuli-Venezia-Giulia Region. In the embedded table, characteristics of out-of-hospital cardiac arrests (OHCAs) with cardiopulmonary resuscitation (CPR) attempted by emergency medical system (EMS) are detailed. BLS, basic life support; ALS, advanced life support. Data in the table are reported as n (%) or median [IQR]. §p < 0.05 and §§p < 0.01 vs. province 2; #p < 0.05, ##p < 0.01 vs. province 3; °p < 0.05 and °°p < 0.01 vs. province 4. Some cardiac arrest location was recorded as “other” or “unknown/not recorded” in the registry, which accounts for the missing percentage in the total.

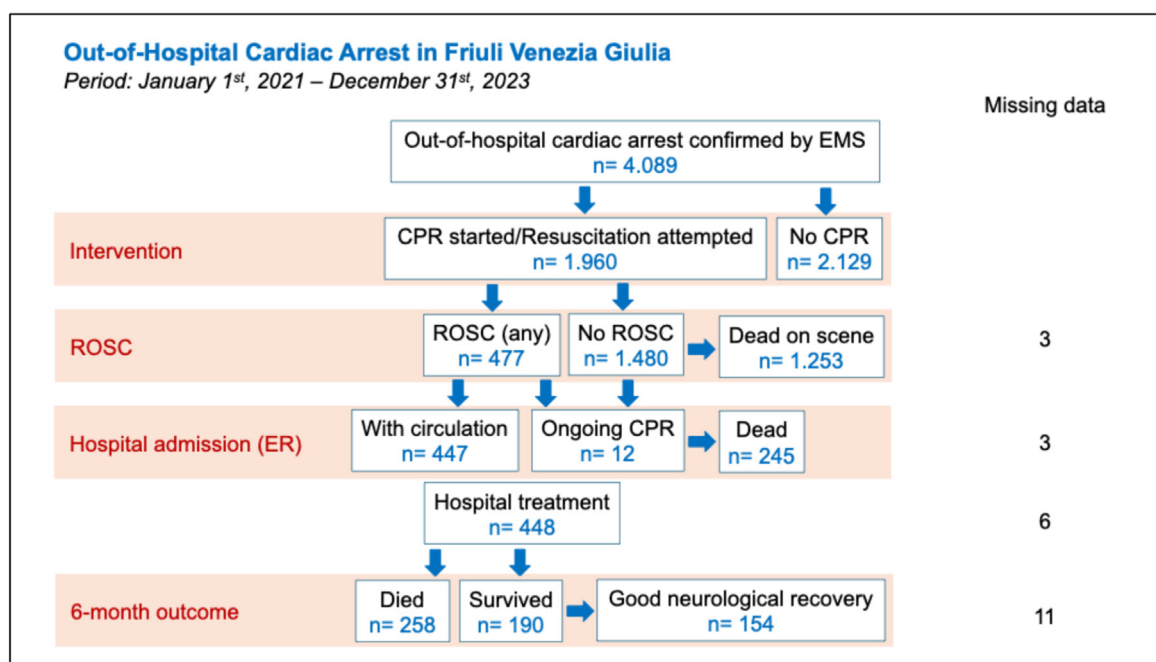


Fig. 2 – Flow-chart with numbers of out-of-hospital cardiac arrest (OHCA) in Friuli-Venezia-Giulia Region. CPR, cardiopulmonary resuscitation; EMS, emergency medical system; ER, emergency room; ROSC, return of spontaneous circulation.

resuscitation was not initiated because deemed futile or inappropriate (i.e. obvious signs of death, terminal illness, injuries incompatible with life), as per EMS documentation.

EMS resuscitation manoeuvres were attempted in 1960 OHCA (48%), with an incidence of 54 per 100,000 inhabitants per year. Characteristics of these EMS-treated OHCA differed significantly across the 4 provinces, as detailed in Fig. 1 and Table 1. Of these patients, 68% were male, and the median age was 73 years. A cardiac aetiology was present in most cases, but a presenting shockable rhythm was registered only in 22% of patients. Most cardiac arrests (77%) occurred at home and more than 80% of all events was witnessed. Overall median no-flow time was 5 min, except for province 3 where the no-flow duration was double ($p < 0.05$).

2024 Utstein OHCA flowcharts for system effectiveness (attempted resuscitation and Utstein comparator)⁹ are reported in the supplemental material (Supplemental Fig. 1 and 2).

Bystander BLS was provided immediately in 67.2% of cases. Prior to EMS arrival, BLS was performed with telephone support from the dispatching centre in 68% of patients with chest compression-only CPR being instructed in almost all cases (Table 1). The urban province 1 had the lowest rates of both bystander CPR and telephone-assisted CPR ($p < 0.05$ vs. province 2). An AED was used by bystanders prior to EMS arrival only in 4.3% of patients. EMS response time, defined as the interval from the emergency call to arrival to the patient, had a median of 11 min and was significantly shorter in the urban province 1 ($p < 0.01$ vs. other provinces). Upon arrival, EMS personnel subjectively rated bystander CPR as high quality in 30% of cases. Overall, the median duration of resuscitation manoeuvres was 29 min.

Bag-mask ventilation was performed in 25% of patients, while 75% received advanced airway management, with the endotracheal (ET) tube being the preferred airway in more than half of events. Mechanical chest compression devices were used in 41% of cases.

Twenty-two per cent of patients survived till hospital admission, while 9.7% were alive at 6 months and 7.6% had a good neurological recovery (Table 2). Province 1 reported significantly better outcomes compared to the other provinces ($p < 0.05$).

Longer CPR durations were associated with lower rates of survival to hospital admission and long-term survival ($p < 0.01$, Fig. 3). Univariable regression showed that younger age, male gender, public location, witnessed cardiac arrest, public AED use, and bystander CPR were all associated with better outcomes (Table 3). Notably, bystander CPR was linked to better short- and long-term survival only when performed with high quality.

At the multivariable regression, variables independently associated with survival to hospital admission, long-term survival, and good neurological recovery, included younger age, OHCA occurring in a public location, presenting shockable rhythm, shorter CPR duration, and lower adrenaline dose (Table 4). Female gender, high-quality bystander CPR, use of a mechanical device, and placement of an advanced airway were associated with increased odds for short-term survival but were not linked to long-term outcomes.

Discussion

This study provides a comprehensive overview of OHCA events with resuscitation attempted by EMS in the FVG region over a three-year period, offering insights into the incidence, resuscitation practices, and outcomes. The results highlight critical factors influencing survival and neurological recovery, underscoring the importance of timely and high-quality CPR interventions. The incidence of OHCA in FVG (i.e. 113 per 100,000 inhabitants per year) was aligned with global trends, together with the predominance of male patients, median age, and proportion of primary cardiac aetiology.^{1,2} Encouragingly, the rate of bystander CPR in FVG was overall high (i.e.

Table 1 – Cardiopulmonary resuscitation characteristics and outcomes in EMS treated out-of-hospital cardiac arrest in the Friuli-Venezia-Giulia Region.

	All <i>n</i> = 1960	Province 1 <i>n</i> = 351	Province 2 <i>n</i> = 866	Province 3 <i>n</i> = 427	Province 4 <i>n</i> = 316
Bystander CPR	1,313 (67.2)	215 (61.3)	602 (69.5)*	278 (65.4)	218 (69.6)
Telephone-CPR	1,325 (68.4)	217 (62)	604 (70.9)*	291 (68.6)	213 (68.5)
CC:V CPR	7 (0.4)	2 (0.6)	5 (0.6)	0 (0)	0 (0)
CCO CPR	1,308 (67)	214 (61)	598 (69.1)*	277 (65.2)	219 (70)
High-quality bystander CPR	573 (30.1)	101 (28.9)	254 (30.2)	188 (44.8)*§	30 (10.2)*##
Bystander AED application	84 (4.3)	20 (5.7)	47 (5.4)	8 (1.9)*§	9 (2.9)
EMS arrival time, min	11 [9–16]	8 [6–11]	12 [9–15]**	12 [9–16]**	11 [8–14]**§§##
CPR duration, min	29 [20–40]	23 [14–33]	29 [18–40]**	28 [18–39]**	35 [18–48] **§§##
Number of defibrillations	2 [1–4]	2 [1–4]	2 [1–5]	2 [1–5]	2 [1–5]
Bag-mask ventilation	495 (25.4)	59 (16.8)	289 (33.7)*	72 (16.9)§	75 (23.7)§
Advanced airway	1,455 (74.6)	292 (83.2)	568 (66.3)*	354 (83.1)§	241 (76.3)§
Laryngeal mask	55 (2.8)	26 (7.4)	27 (3.2)*	0 (0)*§	2 (0.6)*
Laryngeal tube	369 (18.9)	1 (0.3)	163 (19)*	137 (32.2)*§	68 (21.5)*#
Endotracheal intubation	1,031 (52.9)	265 (75.5)	378 (44.1)*	217 (50.9)*	171 (54.1)*§
Mechanical CPR	806 (41.3)	135 (38.5)	275 (31.9)	331 (77.7)*§	65 (20.8)*§#
Adrenaline, yes	1,848 (96.4)	323 (94.2)	814 (96.4)	404 (96.7)	307 (98.4)*
Adrenaline dose, mg	3 [2–5]	3 [1–4]	4 [2–6]**	4 [2–6]**§	5 [3–7]**
Amiodarone, yes	295 (15.1)	65 (18.6)	131 (15.2)	65 (15.3)	34 (10.9)*

Data are reported as number (percentage) or median [IQR].

AED, automated external defibrillation; CCO, chest compression only; CC:V, chest compression and ventilation; CPR, cardiopulmonary resuscitation; Ems, emergency medical service.

* $p < 0.05$ and ** $p < 0.01$ vs. province 1; § $p < 0.05$ and §§ $p < 0.01$ vs. province 2; # $p < 0.05$ and ## $p < 0.01$ vs. province 3.

Table 2 – Survival and long-term outcomes in EMS treated out-of-hospital cardiac arrest in the Friuli-Venezia-Giulia Region.

	All <i>n</i> = 1,960	Province 1 <i>n</i> = 351	Province 2 <i>n</i> = 866	Province 3 <i>n</i> = 427	Province 4 <i>n</i> = 316
Survival to hospital admission	448 (22.9)	128 (36.5)	186 (21.6)*	93 (21.9)*	41 (13)*§#
6-month survival	190 (9.7)	59 (16.8)	82 (9.6)*	37 (8.7)*	12 (3.8)*§#
6-month good neurological outcome	154 (7.9)	37 (10.6)	74 (8.6)	34 (8.0)	9 (2.8)*§

Data are reported as number (percentage).

**p* < 0.05 vs. province 1; §*p* < 0.05 vs. province 2; #*p* < 0.05 vs. province 3.

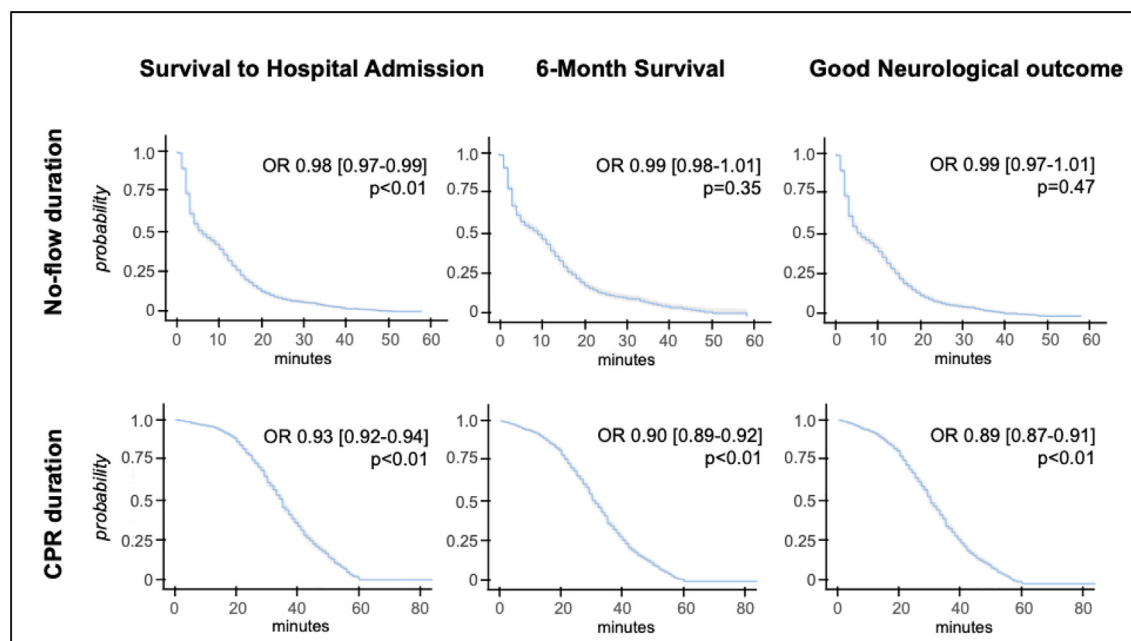


Fig. 3 – Impact of no-flow and cardiopulmonary resuscitation (CPR) duration on survival to hospital admission, 6-month survival, and good neurological recovery (cerebral performance category 1–2), in the emergency medical system-treated out-of-hospital cardiac arrest in Friuli-Venezia-Giulia Region. No-flow time in unwitnessed cardiac arrests was defined as the estimated time from presumed collapse (or last seen well) to initiation of CPR.

67%) and exceeded the European average of 58%, although public AED use (i.e. 4.9%) fitted in the lowest range reported in Europe.^{1,2} Long-term outcomes in FVG (i.e. six-month survival stood at 9.7%, with 7.6% of patients achieving good neurological recovery) were on par with international benchmarks. Indeed, The European Registry of Cardiac Arrest (EuReCa)-ONE study of 10,682 OHCA patients in 27 countries reported a comparable survival rate of 10%, subsequently confirmed by the larger EuReCa-TWO study, that registered an 8% survival.¹

Significant variability in EMS interventions was however observed across the different FVG provinces. Thus, province 1, characterised by a high-density urban area of only 212 km² (i.e. 2–20-fold smaller than the other provinces), had the shortest EMS arrival time and CPR duration in the region. These geographical and demographic features, together with the highest rate of shockable presenting rhythms, witnessed events, and public access defibrillation, likely accounted for the highest percentage of long-term survival with neurological recovery observed in this province. Province 4, instead, was characterised by the lowest rate of survival and good neurological outcome compared to the others, because of multifactorial reasons.

Thus, the presence in this province of the same number of ALS vehicles as province 1, despite a 2-fold larger area with a 4-fold greater population density, might have prolonged the EMS interventions, leading to a long CPR duration. In addition, lesser shockable rhythms, more events occurring at home, together with low public AED use, were also observed in this province compared to the others.^{2,10,11} Province 3, with substantial rural and mountainous terrain, showed the longest no-flow times but relatively favourable outcomes, likely due to effective bystander CPR and aggressive ALS interventions, including high rates of mechanical CPR and airway management (in 77% and 83% of patients, respectively). Finally, the accessibility limits of province 2, the largest and most populated province, encompassing also a vast alpine area, were counteracted by the large number of ALS vehicles available, the high rate of bystander CPR and telephone-assisted CPR with public AED use, such that final outcomes were ultimately not affected.

Italian data on OHCA are incomplete and fragmented due to the lack of a national registry.^{12,13} Nevertheless, available studies suggest an average incidence of 84 events per 100,000 inhabitants annually, and a long-term survival of 9%.^{14,15} This study, derived

Table 3 – Univariable logistic regression for short and long-term outcomes in EMS-treated out-of-hospital cardiac arrest in Friuli-Venezia-Giulia Region.

	Survival to hospital admission		6-month survival		Good neurological recovery	
	OR	95%CI	OR	95%CI	OR	95%CI
Age > 75 year	0.52	0.41–0.66	0.37	0.25–0.55	0.40	0.26–0.61
Gender, female	1.12	0.88–1.43	0.64	0.43–0.96	0.57	0.36–0.91
Public location, yes	3.21	2.41–4.27	5.46	3.74–7.97	4.70	3.11–7.10
Non-shockable rhythm, yes	0.26	0.20–0.33	0.08	0.06–0.12	0.05	0.03–0.09
Witnessed, yes	1.65	1.20–2.26	2.52	1.40–4.52	2.95	1.48–5.91
Bystander CPR, yes	2.20	1.72–2.83	2.56	1.79–3.66	2.59	1.75–3.83
Telephone-CPR, yes	0.84	0.66–1.08	0.85	0.58–1.23	0.85	0.57–1.29
Poor-quality bystander CPR	0.80	0.57–1.13	0.73	0.40–1.33	0.71	0.36–1.41
High-quality bystander CPR	2.33	1.69–3.22	3.14	1.85–5.31	3.27	1.81–5.89
Public AED use, yes	1.70	0.96–2.99	3.04	1.55–5.92	2.66	1.26–5.61
No-flow time > 5 min	0.94	0.74–1.18	0.84	0.59–1.20	0.81	0.55–1.20
EMS arrival > 11 min	0.59	0.47–0.76	0.44	0.29–0.65	0.44	0.28–0.68
CPR duration > 30 min	0.18	0.14–0.24	0.06	0.03–0.13	0.04	0.01–0.11
Mechanical CPR, yes	1.44	1.14–1.81	0.58	0.39–0.85	0.50	0.32–0.77
Advanced airway, yes	3.70	2.50–5.49	1.32	0.73–2.40	0.98	0.51–1.87
Adrenaline > 3 mg	0.48	0.38–0.61	0.13	0.08–0.22	0.08	0.04–0.16
Province						
#4 vs. #1	0.34	0.23–0.50	0.20	0.10–0.39	0.24	0.11–0.52
#3 vs. #1	0.55	0.39–0.77	0.38	0.23–0.63	0.59	0.34–1.02
#2 vs. #1	0.41	0.30–0.56	0.42	0.27–0.64	0.63	0.39–1.01
Year						
2022 vs. 2021	1.10	0.83–1.46	2.25	1.41–3.58	3.06	1.77–5.30
2021 vs. 2021	1.07	0.81–1.42	1.70	1.05–2.76	2.17	1.23–3.84

OR, odds ratio; AED, automated external defibrillator; CPR, cardiopulmonary resuscitation; EMS, emergency medical system.

Table 4 – Multivariable logistic regression for short and long-term outcomes in EMS-treated out-of-hospital cardiac arrest in Friuli-Venezia-Giulia Region.

	Survival to hospital admission		6-month survival		Good neurological recovery	
	adjOR	95%CI	adjOR	95%CI	adjOR	95%CI
Age > 75 year	0.57	0.42–0.76	0.41	0.25–0.68	0.44	0.25–0.78
Gender, female	1.45	1.07–1.96	0.96	0.56–1.64	0.91	0.49–1.70
Public location, yes	2.18	1.53–3.11	3.59	2.11–6.11	3.09	1.68–5.71
Non-shockable rhythm, yes	0.35	0.25–0.49	0.09	0.06–0.16	0.06	0.03–0.11
Witnessed, yes	1.39	0.92–2.01	1.43	0.66–3.13	1.72	0.69–4.32
Bystander CPR						
poor-quality vs. no bystander CPR	1.02	0.65–1.59	0.80	0.36–1.81	0.59	0.23–1.50
high-quality vs. no bystander CPR	2.41	1.55–3.74	1.91	0.90–4.09	1.33	0.56–3.18
Public AED use, yes	0.76	0.34–1.70	0.97	0.32–2.93	0.73	0.22–2.47
No-flow time > 5 min	1.13	0.81–1.58	1.12	0.66–1.93	0.94	0.51–1.75
EMS arrival > 11 min	0.77	0.57–1.04	0.66	0.39–1.12	0.59	0.32–1.10
CPR duration > 30 min	0.18	0.13–0.25	0.13	0.06–0.29	0.10	0.03–0.31
Mechanical CPR, yes	1.69	1.21–2.36	0.57	0.32–1.05	0.42	0.21–0.84
Advanced airway, yes	3.85	2.58–5.72	1.31	0.73–2.36	0.95	0.50–1.79
Adrenaline > 3 mg	0.44	0.32–0.60	0.16	0.08–0.32	0.10	0.04–0.25
Province						
#4 vs. #1	0.64	0.40–1.02	0.57	0.24–1.32	0.97	0.35–2.75
#3 vs. #1	0.54	0.34–0.84	0.58	0.28–1.23	1.60	0.68–3.79
#2 vs. #1	0.60	0.41–0.89	0.62	0.34–1.12	1.28	0.63–2.57
Year						
2022 vs. 2021	0.72	0.51–1.02	1.72	0.94–3.13	2.58	1.26–5.30
2023 vs. 2021	0.70	0.49–1.00	1.03	0.54–1.94	1.24	0.58–2.66

adjOR, adjusted odds ratio; AED, automated external defibrillator; CPR, cardiopulmonary resuscitation; EMS, emergency medical system.

from one of the few regional registries, further supports the gathering of knowledge on national OHCA epidemiology. Another large OHCA registry in Italy covering part of the north-western Lombardia region reported similar population demographics than FVG, but a 2-fold greater incidence of presenting shockable rhythms (i.e. 50% vs. 22%) and approximately half of bystander-initiated CPR and a DA-CPR (i.e. 38.5% vs. 67% and 34.5% vs. 68%, respectively). Similarly to FVG, the public access defibrillation in Lombardia was used only in a minor part of the events (i.e. 4.7%). This gap highlights an urgent need for increased AED availability and public education campaigns, as recommended by international guidelines.¹⁰ Indeed, through registries like the present one, EMSs may have the capability to evaluate their system performance and target key areas of improvement.¹⁰

Effective bystander CPR is a key determinant of OHCA outcome, potentially doubling or tripling the chances of survival.^{10,16–20} DA-CPR also plays a crucial role in increasing bystander engagement, such that dispatch centres have been recommended to have in place systems to provide telephone instructions for OHCA.^{21–24} Based on a recent national survey,²⁵ nearly all EMS dispatch centres in Italy declared to provide telephone-CPR; FVG data confirmed a strong performance in this regard, with a DA-CPR performed in most cases (i.e. 68%).^{10,11} However, while bystander CPR was positively associated with survival at the univariable regression, it did not show any impact on outcomes at the multivariable analysis. This result is likely explained by the low rate (i.e. 30%) of high-quality CPR provided by bystanders, as evaluated by the EMS personnel when arrived at the scene. The provision of high-quality chest compression is critical to re-establish threshold levels of coronary and cerebral perfusion during CPR and this underscores the need for expanded community CPR training and improved dispatcher coaching to optimise DA-CPR practice and quality.^{21,26–28} In our population, only high-quality bystander CPR was associated with both short- and long-term survival in the univariable model, whereas poor-quality bystander CPR had no significant impact on outcomes.

Airway management is another critical component of resuscitation, with an advanced airway usually placed to protect the airway and provide optimal oxygenation.^{21,29} Despite several inconclusive studies,^{30,31} some evidence has suggested that an advanced airway during CPR may improve both short and long-term survival.^{30,32–34} In addition, airway management with an ET tube by physician-staffed EMS might be the optimal airway strategy to improve outcomes in specific settings.^{32–34} Thus, a recent multicentre-cohort study enrolling more than 16,000 patients in Japan found that pre-hospital advanced airway was significantly associated with increased neurological outcome when managed under the presence of a physician.³⁴ In FVG, the response to an OHCA is based on a physician (anaesthetist)-led ALS team, skilled in the tracheal intubation manoeuvres, or to an emergency nurse-based team, trained in placing a laryngeal tube. For this, the rate of advanced airway establishment was impressively high (75%); however, this treatment was associated only with short-term survival but not with long-term one.

Survival strongly correlates with short no-flow time and early initiation of resuscitation manoeuvres, together with rapid EMS response time.^{16,17,35–38} Similarly, a short CPR duration has been associated with improved OHCA outcomes, emphasising the critical role of rapid and effective resuscitation. Although some studies have shown that patients might benefit from longer resuscitation periods,³⁹ there is solid evidence highlighting that survival rate decreases as CPR duration increases, likely reflecting delayed or unsuccessful

ROSC, rather than a direct cause of poor outcome.^{40,41} In our study, the median EMS response time of 11 min and the median resuscitation duration of 29 min, although in line with other European data, reflected challenges in providing timely interventions, particularly in rural or remote areas.

Differences in OHCA outcomes between men and women have been reported in previous studies, although it has not been conclusively established whether these differences are due to varying risk factors or disparities in the quality of treatment.^{42–46} The above results have been explained, in part, by the low chance of women to receive bystander CPR and having a public access AED applied compared to men.^{45,47–49} Our data aligns with the above evidence, as female patients had lower odds for long-term survival and neurological recovery compared to men. Further investigations are required to clarify these findings and propose solutions to improve bystander intervention in the instance of cardiac arrest in a woman.

We recognise several limitations in the interpretation of our findings. First, the regional registry was developed using the Utstein Registry as a reference,⁸ but it does not capture all the variables outlined in the Utstein template. Additionally, certain variables have been adapted to suit our specific clinical and data collection context, which may limit direct comparability with fully Utstein-compliant registries. Nevertheless, the registry records important additional data such as long-term survival and neurological recovery that may offer clinically relevant information. Second, the retrospective design and the potential for variability in data collection across provinces. However, data were prospectively acquired, and the medical officers assured their completeness and quality; in addition, we were able to adjust the model to overcome any potential biases. Third, the health conditions, previous diseases and lifestyles of the patients remained unknown. Additionally, factors such as socioeconomic status and access to care were not explored but may have influenced outcomes. Nevertheless, the FVG OHCA registry represents the only registry covering a whole Region in Italy and by encompassing different geographical areas (i.e. urban, rural, and mountainous) it offers a valid representation of the epidemiology of OHCA and EMS organisation, that can be translated to the whole country. The findings of this study emphasise the critical role of early and high-quality interventions to improve OHCA outcomes. While objective measures of bystander-performed CPR quality were not available, the assessments conducted by EMS personnel, predominantly anaesthetists, are supported by their clinical expertise in evaluating CPR performance. Thus, despite the inherent subjectivity, the data offer valuable and meaningful insights into the quality of bystander intervention. Future efforts should focus on strategies to enhance the quality of bystander CPR, public AED use, and optimise EMS response times to further improve outcomes in the region. These results provide a valuable foundation for public health initiatives and policy-making aimed at reducing the burden of cardiac arrest.

Conclusions

This study provides comprehensive insights into OHCA epidemiology in the FVG region and emphasises the importance of early intervention, high-quality bystander CPR, and specialised prehospital care to improve outcomes. Policy initiatives should target CPR training, AED accessibility, and EMS optimisation while addressing regional disparities to enhance survival.

CRedit authorship contribution statement

Carlo Pegani: Writing – review & editing, Data curation, Conceptualization. **Alberto Peratoner:** Writing – review & editing, Supervision, Resources, Conceptualization. **Manila Andrian:** Validation, Data curation. **Laura Magagnin:** Validation, Data curation. **Luca Gambolò:** Methodology, Formal analysis. **Alice Pravisan:** Writing – original draft, Validation, Conceptualization. **Michele Zuliani:** Validation, Data curation. **Serena Rakar:** Supervision, Resources. **Cristina Lutman:** Investigation, Data curation. **Alessio Della Mattia:** Investigation, Data curation. **Katya Fabretto:** Validation, Supervision. **Annarita Tullio:** Methodology, Formal analysis. **Giuseppe Stirparo:** Writing – original draft, Formal analysis. **Erik Roman Pog-nuz:** Writing – original draft, Supervision. **Giuseppe Ristagno:** Writing – original draft, Methodology, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

Collaborators of the FVG-OHCA Registry network: Stefano Savorgnani, Antonella Petrei, Benedetta Cordova, Giulio Trillò, Eliana Franceschino, Michele Contadini, Eugenio Montagner, Davide Caggegi, Amato De Monte, Vincenzo Mione.

Funding

This study was funded in part by Italian Ministry of Health-Current research IRCCS.

Appendix 1

Collaborators: The FVG-OHCA Registry network: Stefano Savorgnani, Antonella Petrei, Benedetta Cordova, Giulio Trillò, Eliana Franceschino, Michele Contadini, Eugenio Montagner, Davide Caggegi, Amato De Monte, Vincenzo Mione.

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.resplu.2025.101000>.

Author details

the FVG-OHCA registry network¹ ^aPrehospital Emergency Medical Service, Department of Anaesthesia and Intensive Care, Cattinara Hospital, University of Trieste, Trieste, Italy ^bRegional Emergency Medical Dispatch Centre, Central Regional Health Agency, Udine, Italy ^cPrehospital Emergency Medical Service, Santa Maria degli Angeli, Pordenone, Italy ^dSIMED-Società Italiana di Medicina e Divulgazione Scientifica, Parma, Italy ^eIntensive Care Unit, Santa Maria degli Angeli, Pordenone, Italy ^fDepartment of

Anaesthesia and Intensive Care, Health Integrated Agency of Friuli Centrale, Udine, Italy ^gCardiovascular Department, Cattinara Hospital, University of Trieste, Trieste, Italy ^hCardiovascular Department, Health Integrated Agency of Friuli Centrale, Udine, Italy ⁱCardiovascular Department, Santa Maria degli Angeli, Pordenone, Italy ^jDepartment of Medicine (DMED), University of Udine, Udine, Italy ^kDepartment of Medical Science, Intensive Care Unit, University Hospital of Cattinara, Department of Anaesthesia, University of Trieste, Trieste, Italy ^lDepartment of Pathophysiology and Transplantation, University of Milan, Italy ^mDepartment of Anaesthesiology, Intensive Care and Emergency, Fondazione IRCCS Ca' Granda Ospedale Maggiore Policlinico, Milan, Italy

REFERENCES

- Gräsner JT, Wnent J, Herlitz J, et al. Survival after out-of-hospital cardiac arrest in Europe - results of the EuReCa TWO study. *Resuscitation* 2020;148:218–26.
- Gräsner JT, Herlitz J, Tjelmeland IBM, et al. European resuscitation council guidelines 2021: epidemiology of cardiac arrest in Europe. *Resuscitation* 2021;161:61–79.
- Nishiyama C, Kiguchi T, Okubo M, et al. Three-year trends in out-of-hospital cardiac arrest across the world: second report from the International Liaison Committee on Resuscitation (ILCOR). *Resuscitation* 2023;186:109757.
- Møller SG, Wissenberg M, Møller-Hansen S, et al. Regional variation in out-of-hospital cardiac arrest: incidence and survival – A nationwide study of regions in Denmark. *Resuscitation* 2020;148:191–9.
- Empana JP, Blom MT, Böttiger BW, et al. Determinants of occurrence and survival after sudden cardiac arrest – A European perspective: the ESCAPE-NET project. *Resuscitation* 2018;124:7–13.
- Abir M, Fouche S, Lehrich J, et al. Variation in pre-hospital outcomes after out-of-hospital cardiac arrest in Michigan. *Resuscitation* 2021;158:201–7.
- Siddiqui FJ, McNally B, Gräsner JT, et al. Towards advancing out-of-hospital cardiac arrest (OHCA) registries globally: considerations from the global OHCA registry (GOHCAR) collaborative. *Resusc Plus* 2024;18:100615.
- Jacobs I, Nadkarni V, Bahr J, et al. Cardiac arrest and cardiopulmonary resuscitation outcome reports: update and simplification of the Utstein templates for resuscitation registries. *Resuscitation* 2004;63:233–49.
- Grasner JT, Bray JE, Nolan JP, et al. Cardiac arrest and cardiopulmonary resuscitation outcome reports: 2024 update of the Utstein out-of-hospital cardiac arrest registry template. *Resuscitation* 2024;201:110288.
- Semeraro F, Greif R, Böttiger BW, et al. European resuscitation council guidelines 2021: systems saving lives. *Resuscitation* 2021;161:80–97.
- Olasveengen TM, Semeraro F, Ristagno G, et al. European resuscitation council guidelines 2021: basic life support. *Resuscitation* 2021;161:98–114.
- Ristagno G, Semeraro F, Radeschi G, et al. The “Italian registry of Cardiac arrest - RIAC”, a national achievement to portray the Italian reality and to contribute to the wider European vision by “EuReCa”. *Resuscitation* 2014;85:e193–4.
- Sgueglia AC, Gentile L, Bertuccio P, et al. Out-of-hospital cardiac arrest outcomes' determinants: an Italian retrospective cohort study based on Lombardia CaRe. *Intern Emerg Med* 2024;19:2035–45.
- Scquizzato T, Gamberini L, D'Arrigo S, et al. Incidence, characteristics, and outcome of out-of-hospital cardiac arrest in Italy:

- a systematic review and meta-analysis. *Resusc Plus* 2022;12:100329.
15. Gräsner JT, Lefering R, Koster RW, et al. EuReCa ONE-27 nations, ONE Europe, ONE registry: a prospective one-month analysis of out-of-hospital cardiac arrest outcomes in 27 countries in Europe. *Resuscitation* 2016;105:188–95.
 16. Hsu Y-C, Wu W-T, Huang J-B, et al. Association between prehospital prognostic factors and out-of-hospital cardiac arrest: effect of rural-urban disparities. *Am J Emerg Med* 2021;46:456–61.
 17. Yoshimoto H, Fukui K, Nishimoto Y, et al. Annual improvement trends in resuscitation outcome of patients defibrillated by laypersons after out-of-hospital cardiac arrests and compression-only resuscitation of laypersons. *Resuscitation* 2023;183:109672.
 18. Sondergaard KB, Wissenberg M, Gerdts TA, et al. Bystander cardiopulmonary resuscitation and long-term outcomes in out-of-hospital cardiac arrest according to location of arrest. *Eur Heart J* 2019;40:309–18.
 19. Kragholm K, Wissenberg M, Mortensen RN, et al. Bystander efforts and 1-year outcomes in out-of-hospital cardiac arrest. *N Engl J Med* 2017;376:1737–47.
 20. Pollack RA, Brown SP, Rea T, et al. Impact of bystander automated external defibrillator use on survival and functional outcomes in shockable observed public cardiac arrests. *Circulation* 2018;137:2104–13.
 21. Greif R, Bray JE, Djärv T, et al. 2024 International Consensus on Cardiopulmonary Resuscitation and Emergency Cardiovascular Care science with treatment recommendations: summary from the basic life support; advanced life support; pediatric life support; neonatal life support; education, implementation, and teams; and first aid task forces. *Resuscitation* 2024;205:110414.
 22. Lee SCL, Mao DR, Ng YY, et al. Emergency medical dispatch services across Pan-Asian countries: a web-based survey. *BMC Emerg Med* 2020;20:1.
 23. Tjelmeland IBM, Masterson S, Herlitz J, et al. GL2020 epidemiology group and participating countries. description of emergency medical services, treatment of cardiac arrest patients and cardiac arrest registries in Europe. *scand J trauma resusc. Emerg Med* 2020;28:103.
 24. Beck B, Bray JE, Smith K, et al. Aus-ROC steering committee. Description of the ambulance services participating in the Aus-ROC Australian and New Zealand out-of-hospital cardiac arrest epistery. *Emerg Med Australas* 2016;28:673.
 25. Di Marco S, Tucci R, Tonelli G, et al. Preparedness for telephone dispatch-assisted cardiopulmonary resuscitation in Italy. A National Survey. *Resuscitation* 2020;149:87–8.
 26. Magliocca A, Castagna V, Fornari C, et al. Transthoracic impedance variability to assess quality of chest compression in out-of-hospital cardiac arrest. *Acta Anaesthesiol Scand* 2024;68:556–66.
 27. Ristagno G, Tang W, Chang YT, et al. The quality of chest compressions during cardiopulmonary resuscitation overrides importance of timing of defibrillation. *Chest* 2007;132:70–5.
 28. Lee HS, You K, Jeon JP, Kim C, Kim S. The effect of video-instructed versus audio instructed dispatcher-assisted cardiopulmonary resuscitation on patient outcomes following out of hospital cardiac arrest in Seoul. *Sci Rep* 2021;11:15555.
 29. Soar J, Böttiger BW, Carli P, et al. European resuscitation council guidelines 2021: adult advanced life support. *Resuscitation* 2021;161:115–51.
 30. Bengner JR, Kirby K, Black S, et al. Effect of a strategy of a supraglottic airway device vs tracheal intubation during out-of-hospital cardiac arrest on functional outcome: the AIRWAYS-2 randomized clinical trial. *JAMA* 2018;320:779–91.
 31. Battaglini D, Schiavetti I, Ball L, et al. Association between early airway intervention in the pre-hospital setting and outcomes in out of hospital cardiac arrest patients: a post-hoc analysis of the Target temperature Management-2 (TTM2) trial. *Resuscitation* 2024;203:110390.
 32. Hasegawa K, Hiraide A, Chang Y, Brown DFM. Association of prehospital advanced airway management with neurologic outcome and survival in patients with out-of-hospital cardiac arrest. *JAMA* 2013;309:257.
 33. Behrens NH, Fischer M, Krieger T, et al. Effect of airway management strategies during resuscitation from out-of-hospital cardiac arrest on clinical outcome: a registry-based analysis. *Resuscitation* 2020;152:157–64.
 34. Hatakeyama T, Kiguchi T, Sera T, et al. Pre-hospital airway management and neurological status of patients with out-of-hospital cardiac arrest: a retrospective cohort study. *Resusc Plus* 2023;15:100422.
 35. Ristagno G, Fumagalli F, Russo I, et al. Postresuscitation treatment with argon improves early neurological recovery in a porcine model of cardiac arrest. *Shock* 2014;41:72–8.
 36. Grunau B, Kawano T, Scheuermeyer F, et al. Early advanced life support attendance is associated with improved survival and neurologic outcomes after non-traumatic out-of-hospital cardiac arrest in a tiered prehospital response system. *Resuscitation* 2019;135:137–44.
 37. Moon S, Ryoo HW, Ahn JY, et al. Association of response time interval with neurological outcomes after out-of-hospital cardiac arrest according to bystander CPR. *Am J Emerg Med* 2020;38:1760–6.
 38. Kim DK, Kim TH, Do SS, et al. Impact of crowding in local ambulance demand on call-to-ambulance scene arrival in out-of-hospital cardiac arrest. *Am J Emerg Med* 2022;52:105–9.
 39. Reynolds JC, Grunau BE, Rittenberger JC, et al. Association between duration of resuscitation and favorable outcome after out-of-hospital cardiac arrest. *Circulation* 2016;134:2084–94.
 40. Rea T, Kudenchuk PJ, Sayre MR, et al. Out of hospital cardiac arrest: past, present, and future. *Resuscitation* 2021;165:101–9.
 41. Matsuyama T, Ohta B, Kiyohara K, Kitamura T. Cardiopulmonary resuscitation duration and favorable neurological outcome after out-of-hospital cardiac arrest: a nationwide multicenter observational study in japan (the JAAM-OHCA registry). *Crit Care* 2022;26:120.
 42. Choi HZ, Chang H, Ko SH, Kim MC. Gender effect in survival after out-of-hospital cardiac arrest: a nationwide, population-based, case control propensity score matched study based Korean national cardiac arrest registry. *PLoS One* 2022;17:e0258673.
 43. Blom MT, Oving I, Berdowski J, van Valkengoed IGM, Bardai A, Tan HL. Women have lower chances than men to be resuscitated and survive out-of-hospital cardiac arrest. *Eur Heart J* 2019;40:3824–34.
 44. Malik A, Gewarges M, Pezzutti O, et al. Association between sex and survival after non-traumatic out of hospital cardiac arrest: a systematic review and meta-analysis. *Resuscitation* 2022;179:172–82.
 45. Blewer AL, McGovern SK, Schmicker RH, et al. Gender disparities among adult recipients of bystander cardiopulmonary resuscitation in the public. *Circ Cardiovasc Qual Outcomes* 2018;11:e004710.
 46. Smits RLA, Södergren STF, Folke F, et al. Long-term survival following out-of-hospital cardiac arrest in women and men: influence of comorbidities, social characteristics, and resuscitation characteristics. *Resuscitation* 2024;201:110265.
 47. Liu N, Ning Y, Ong MEH, et al. Gender disparities among adult recipients of layperson bystander cardiopulmonary resuscitation by location of cardiac arrest in Pan-Asian communities: a registry-based study. *EClinicalMedicine* 2022;44:101293.
 48. Munot S, Bray JE, Redfern J, et al. Bystander cardiopulmonary resuscitation differences by sex - the role of arrest recognition. *Resuscitation* 2024;199:110224.
 49. Böckler B, Preisner A, Bathe J, et al. Gender-related differences in adults concerning frequency, survival and treatment quality after out-of-hospital cardiac arrest (OHCA): an observational cohort study from the German resuscitation registry. *Resuscitation* 2024;194:110060.